



PART E

Numeric Analysis

Software (p. 788–789)

CHAPTER 19 Numerics in General

CHAPTER 20 Numeric Linear Algebra

CHAPTER 21 Numerics for ODEs and PDEs

Numeric analysis or briefly **numerics** continues to be one of the fastest growing areas of engineering mathematics. This is a natural trend with the ever greater availability of computing power and global Internet use. Indeed, good software implementation of numerical methods are readily available. Take a look at the *updated* list of **Software** starting on p. 788. It contains software for purchase (commercial software) and software for free download (public-domain software). For convenience, we provide Internet addresses and phone numbers. The software list includes computer algebra systems (CASs), such as *Maple* and *Mathematica*, along with the *Maple Computer Guide*, 10th ed., and *Mathematica Computer Guide*, 10th ed., by E. Kreyszig and E. J. Norminton related to this text that teach you stepwise how to use these computer algebra systems and with complete engineering examples drawn from the text. Furthermore, there is scientific software, such as *IMSL*, *LAPACK* (free download), and scientific calculators with graphic capabilities such as *TI-Nspire*. Note that, although we have listed frequently used quality software, this list is by no means complete.

In your career as an engineer, applied mathematician, or scientist you are likely to use commercially available software or proprietary software, owned by the company you work for, that uses numeric methods to solve engineering problems, such as modeling chemical or biological processes, planning ecologically sound heating systems, or computing trajectories of spacecraft or satellites. For example, one of the collaborators of this book (Herbert Kreyszig) used proprietary software to determine the value of bonds, which amounted to solving higher degree polynomial equations, using numeric methods discussed in Sec. 19.2.

*However, the availability of quality software does not alleviate your effort and responsibility to first **understand** these numerical methods.* Your effort will pay off because, with your mathematical expertise in numerics, you will be able to plan your solution approach, judiciously select and use the appropriate software, judge the quality of software, and, perhaps, even write your own numerics software.

Numerics extends your ability to solve problems that are either difficult or impossible to solve analytically. For example, certain integrals such as error function [see App. 3, formula (35)] or large eigenvalue problems that generate high-degree characteristic polynomials cannot be solved analytically. Numerics is also used to construct approximating polynomials through data points that were obtained from some experiments.

Part E is designed to give you a solid background in numerics. We present many numeric methods as **algorithms**, which give these methods in detailed steps suitable for software implementation on your computer, CAS, or programmable calculator. The first chapter, Chap. 19, covers three main areas. These are general numerics (floating point, rounding errors, etc.), solving equations of the form $f(x) = 0$ (using Newton's method and other methods), interpolation along with methods of numeric integration that make use of it, and differentiation.

Chapter 20 covers the essentials of numeric linear algebra. The chapter breaks into two parts: solving linear systems of equations by methods of Gauss, Doolittle, Cholesky, etc. and solving eigenvalue problems numerically. Chapter 21 again has two themes: solving ordinary differential equations and systems of ordinary differential equations as well as solving partial differential equations.

Numerics is a very active area of research as new methods are invented, existing methods improved and adapted, and old methods—impractical in precomputer times—are rediscovered. A main goal in these activities is the development of well-structured software. And in large-scale work—millions of equations or steps of iterations—even small algorithmic improvements may have a large significant effect on computing time, storage demand, accuracy, and stability.

Remark on Software Use. Part E is designed in such a way as to *allow complete flexibility on the use of CASs, software, or graphing calculators*. The computational requirements range from very little use to heavy use. The choice of computer use is at the discretion of the professor. The material and problem sets (except where clearly indicated such as in CAS Projects, CAS Problems, or CAS Experiments, which can be omitted without loss of continuity) do not require the use of a CAS or software. A scientific calculator perhaps with graphing capabilities is all that is required.

Software

See also <http://www.wiley.com/college/kreyszig/>

The following list will help you if you wish to find software. You may also obtain information on known and new software from websites such as Dr. Dobb's Portal, from articles published by the *American Mathematical Society* (see also its website at www.ams.org), the *Society for Industrial and Applied Mathematics* (SIAM, at www.siam.org), the *Association for Computing Machinery* (ACM, at www.acm.org), or the *Institute of Electrical and Electronics Engineers* (IEEE, at www.ieee.org). Consult also your library, computer science department, or mathematics department.

TI-Nspire. Includes TI-Nspire CAS and programmable graphic calculators. Texas Instruments, Inc., Dallas, TX. Telephone: 1-800-842-2737 or (972) 917-8324; website at www.education.ti.com.

EISPACK. See LAPACK.

GAMS (Guide to Available Mathematical Software). Website at <http://gams.nist.gov>. Online cross-index of software development by NIST.

IMSL (International Mathematical and Statistical Library). Visual Numerics, Inc., Houston, TX. Telephone: 1-800-222-4675 or (713) 784-3131; website at www.vni.com. Mathematical and statistical FORTRAN routines with graphics.

LAPACK. FORTRAN 77 routines for linear algebra. This software package supersedes LINPACK and EISPACK. You can download the routines from www.netlib.org/lapack. The LAPACK User's Guide is available at www.netlib.org.

LINPACK see LAPACK

Maple. Waterloo Maple, Inc., Waterloo, ON, Canada. Telephone: 1-800-267-6583 or (519) 747-2373; website at www.maplesoft.com.

Maple Computer Guide. For Advanced Engineering Mathematics, 10th edition. By E. Kreyszig and E. J. Norminton. John Wiley and Sons, Inc., Hoboken, NJ. Telephone: 1-800-225-5945 or (201) 748-6000.

Mathcad. Parametric Technology Corp. (PTC), Needham, MA. Website at www.ptc.com.

Mathematica. Wolfram Research, Inc., Champaign, IL. Telephone: 1-800-965-3726 or (217) 398-0700; website at www.wolfram.com.

Mathematica Computer Guide. For Advanced Engineering Mathematics, 10th edition. By E. Kreyszig and E. J. Norminton. John Wiley and Sons, Inc., Hoboken, NJ. Telephone: 1-800-225-5945 or (201) 748-6000.

Matlab. The MathWorks, Inc., Natick, MA. Telephone: (508) 647-7000; website at www.mathworks.com.

NAG. Numerical Algorithms Group, Inc., Lisle, IL. Telephone: (630) 971-2337; website at www.nag.com. Numeric routines in FORTRAN 77, FORTRAN 90, and C.

NETLIB. Extensive library of public-domain software. See at www.netlib.org.

NIST. National Institute of Standards and Technology, Gaithersburg, MD. Telephone: (301) 975-6478; website at www.nist.gov. For Mathematical and Computational Science Division telephone: (301) 975-3800. See also <http://math.nist.gov>.

Numerical Recipes. Cambridge University Press, New York, NY. Telephone: 1-800-221-4512 or (212) 924-3900; website at www.cambridge.org/us. Book, 3rd ed. (in C++) see App. 1, Ref. [E25]; source code on CD ROM in C++, which also contains old source code (but not text) for (out of print) 2nd ed. C, FORTRAN 77, FORTRAN 90 as well as source code for (out of print) 1st ed. To order, call office at West Nyack, NY, at 1-800-872-7423 or (845) 353-7500 or online at www.nr.com.

FURTHER SOFTWARE IN STATISTICS. See Part G.